多模态时间序列预测：
对于 文本模态融合 的 有效性与机制 初步探究

汇报人：10242150443 吴晨曦

时间：2026.6.26

研究问题

- **文本模态是否有效? ——if?**
  - 数值baseline：DLinear、PatchTST
- **怎样的文本有效? ——what?**
  - random随机文本 VS LLM-generated自生文本 VS 外源news/report
- **如何融合更有效? ——how?**
  - 线性融合 VS 非线性融合 VS 频域分解融合

对比实验一

对比实验二

对比实验一：数值基线

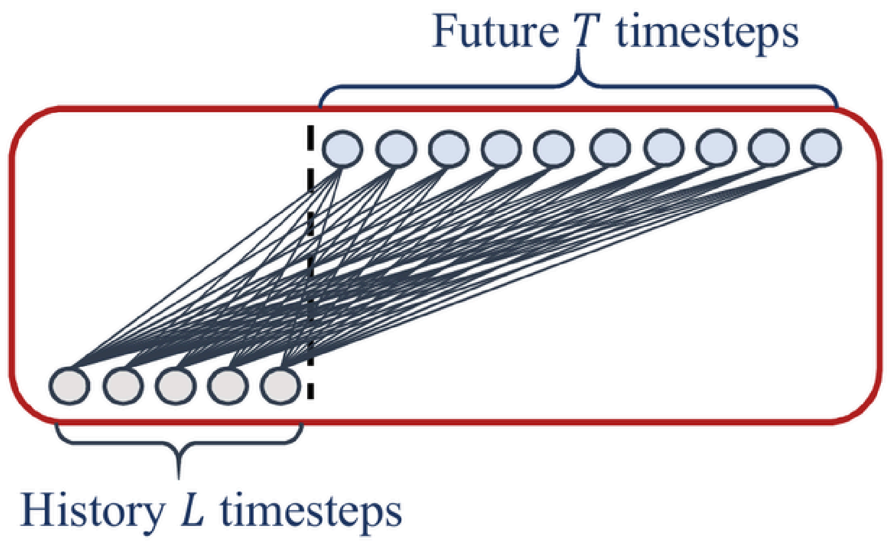

figure1.basic linear model

DLinear：基础线性基线

PatchTST：Transformer-based基线

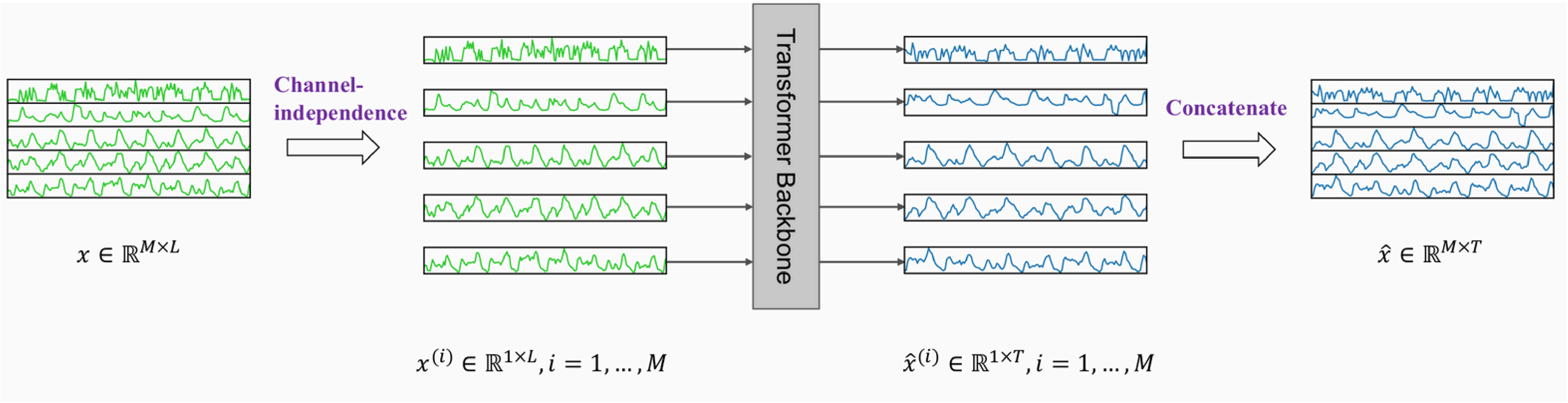

figure2.PatchTST Model Overviewl

对比实验一：基于MM-TSFlib的文本数值基础融合

- 数值预测分支：num_pred = PatchTST/DLinear(batch_x)
- 文本预测分支：
  - text_signal = pool(MLP(LLM_embedding(batch_text))) 文本信号
  - prompt_y = norm(text_signal) + prior_y

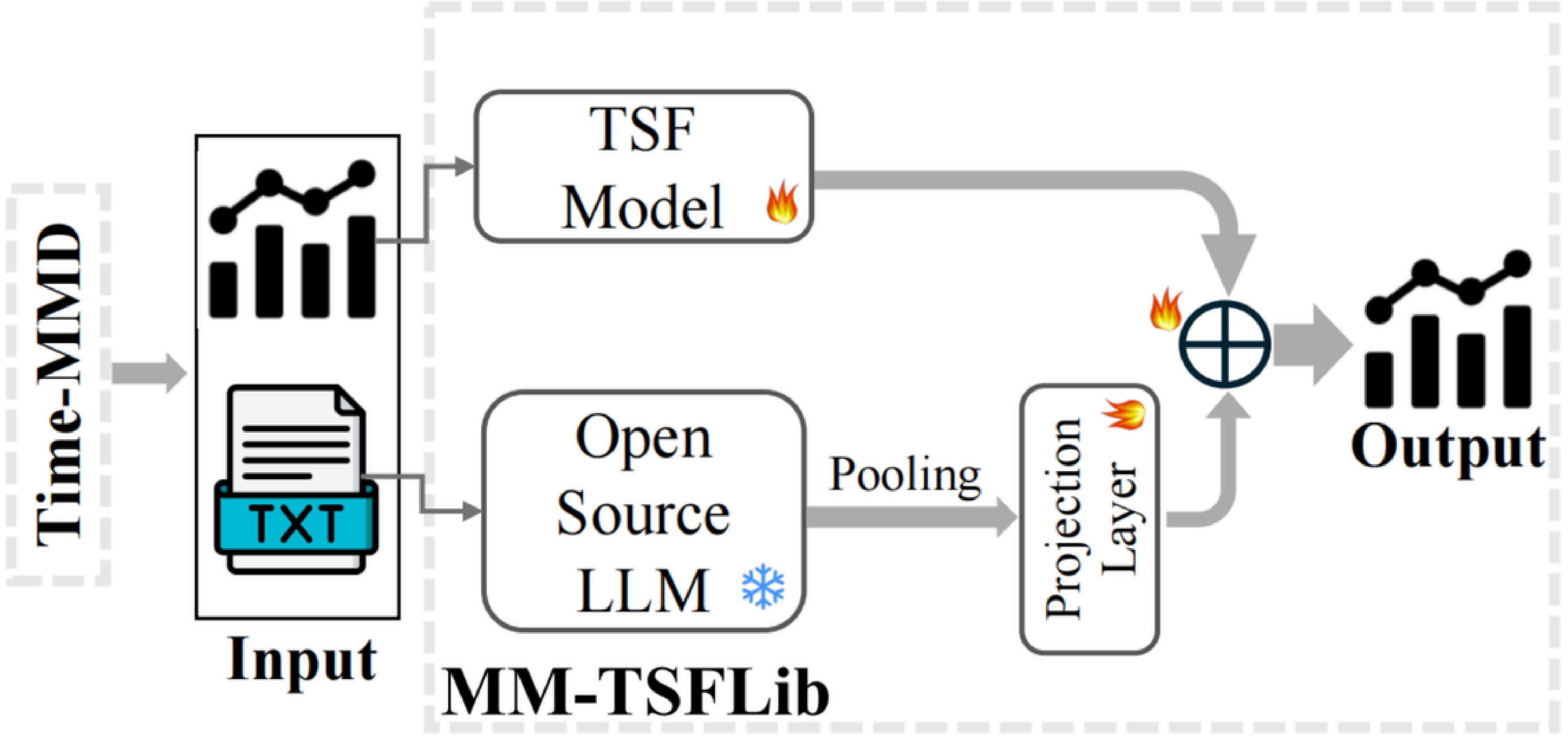

figure3. MM-TSFLib Overviewl

对比实验一：数据集选择与构建

1.Time-MMD数据集

- **多领域多模态数据集**
- 文本：通过 **报告**（Report）与 **网络搜索**（Web Search） 双源采集
- **时间对齐：** start_date,end_date
- **泄露控制：** max(text_end_time) <= input boundary

2.随机文本数据集

- 英文字母、空格和标点中**随机采样**
- **生成长度**与真实文本长度类似

3.LLM生成文本数据集

- “Use only the **historical sequence** and **variable meanings** supplied in the prompt.
- <u>**Do not use**</u> future values, future events, hidden labels, external search, or prior knowledge not grounded in the prompt.
- Return valid JSON with exactly these string keys: \"**fact**\" and \"**pred**\"."

对比实验一：结果展示与分析

backbone：DLinear

dataset	pred_lens	metric	DLinear Numeric	DLinear Random	DLinear LLM	DLinear External
Algriculture	12/24/36/48	MSE	1.450	1.418 (-0.032)	1.420 (-0.030)	**1.417** (-0.033)
Algriculture	12/24/36/48	MAE	0.833	0.819 (-0.013)	0.820 (-0.013)	**0.819** (-0.013)
Climate	12/24/36/48	MSE	1.187	1.112 (-0.075)	**1.112** (-0.075)	1.112 (-0.075)
Climate	12/24/36/48	MAE	0.862	0.835 (-0.027)	**0.835** (-0.028)	0.835 (-0.027)
Economy	12/24/36	MSE	0.555	0.536 (-0.019)	**0.532** (-0.023)	0.536 (-0.019)
Economy	12/24/36	MAE	0.612	0.604 (-0.008)	**0.601** (-0.011)	0.604 (-0.008)
Energy	12/24/36/48	MSE	0.347	**0.318** (-0.029)	0.318 (-0.029)	0.318 (-0.029)
Energy	12/24/36/48	MAE	0.453	0.418 (-0.035)	**0.418** (-0.035)	0.418 (-0.035)
Public_Health	12/24/36/48	MSE	2.441	1.934 (-0.507)	**1.930** (-0.511)	1.934 (-0.507)
Public_Health	12/24/36/48	MAE	1.148	0.961 (-0.187)	**0.959** (-0.188)	0.961 (-0.187)
Security	12/24	MSE	73.639	73.198 (-0.441)	**73.195** (-0.444)	**73.195** (-0.444)
Security	12/24	MAE	4.237	4.137 (-0.100)	**4.137** (-0.100)	**4.137** (-0.100)
SocialGood	12/24/36/48	MSE	1.761	**1.527** (-0.234)	1.529 (-0.233)	1.528 (-0.234)
SocialGood	12/24/36/48	MAE	1.004	0.931 (-0.073)	**0.929** (-0.075)	0.931 (-0.073)
Traffic	12/24/36/48	MSE	0.323	0.295 (-0.027)	0.295 (-0.028)	**0.294** (-0.029)
Traffic	12/24/36/48	MAE	0.439	0.428 (-0.011)	**0.428** (-0.011)	0.429 (-0.010)

对比实验一：结果展示与分析

backbone：PatchTST

dataset	pred_lens	metric	PatchTST Numeric	PatchTST Random	PatchTST LLM	PatchTST External
Algriculture	12/24/36/48	MSE	1.096	**1.030** (-0.065)	1.031 (-0.065)	1.031 (-0.065)
Algriculture	12/24/36/48	MAE	0.719	0.704 (-0.015)	0.704 (-0.015)	**0.704** (-0.015)
Climate	12/24/36/48	MSE	1.176	1.059 (-0.117)	**1.057** (-0.119)	1.060 (-0.116)
Climate	12/24/36/48	MAE	0.855	0.816 (-0.039)	**0.816** (-0.039)	0.817 (-0.038)
Economy	12/24/36	MSE	0.318	**0.303** (-0.015)	0.304 (-0.014)	0.303 (-0.015)
Economy	12/24/36	MAE	0.449	0.435 (-0.014)	0.436 (-0.014)	**0.435** (-0.015)
Energy	12/24/36/48	MSE	**0.253**	0.254 (+0.002)	0.255 (+0.002)	0.254 (+0.001)
Energy	12/24/36/48	MAE	**0.362**	0.366 (+0.005)	0.367 (+0.005)	0.366 (+0.004)
Public_Health	12/24/36/48	MSE	1.744	1.590 (-0.154)	1.589 (-0.155)	**1.589** (-0.156)
Public_Health	12/24/36/48	MAE	0.892	**0.835** (-0.056)	0.837 (-0.055)	0.836 (-0.056)
Security	12/24	MSE	83.160	81.877 (-1.283)	**81.241** (-1.919)	81.804 (-1.356)
Security	12/24	MAE	5.101	5.001 (-0.099)	**4.964** (-0.137)	4.996 (-0.104)
SocialGood	12/24/36/48	MSE	**1.514**	1.518 (+0.004)	1.531 (+0.017)	1.515 (+0.001)
SocialGood	12/24/36/48	MAE	0.802	0.794 (-0.008)	**0.790** (-0.012)	0.791 (-0.011)
Traffic	12/24/36/48	MSE	0.296	**0.283** (-0.013)	0.284 (-0.012)	0.283 (-0.013)
Traffic	12/24/36/48	MAE	0.366	**0.348** (-0.018)	0.349 (-0.017)	0.348 (-0.018)

对比实验二：融合机制探究

1.Linear Fusion 基础线性融合

数值　　文本

pred_result= w * num + (1-w）* text

2.Nonlinear Fusion 非线性融合

Z = concat(Y_num, Y_text,Y_text - Y_num)

G　= sigmoid(**MLP_gate**(Z))

pred_result=(1-G) * Y_num + G * Y_text

3.Frequency Decomposition Fusion 频域分解融合

数值 → 数值低频 / 数值中频 / 数值高频

文本 → 文本低频 / 文本中频 / 文本高频

weight

pred_result

对比实验二：融合机制探究

dataset	pred_lens	metric	DLinear Linear	DLinear Nonlinear	DLinear Frequency	PatchTST Linear	PatchTST Nonlinear	PatchTST Frequency
Algriculture	12/24/36/48	MSE	**1.417**	1.417 (+0.000)	1.463 (+0.046)	1.031	**1.031** (-0.000)	1.188 (+0.157)
Algriculture	12/24/36/48	MAE	0.819	**0.819** (-0.000)	0.829 (+0.010)	0.704	**0.702** (-0.002)	0.740 (+0.036)
Climate	12/24/36/48	MSE	1.112	1.106 (-0.006)	**1.093** (-0.019)	1.060	1.058 (-0.002)	**1.039** (-0.021)
Climate	12/24/36/48	MAE	0.835	0.833 (-0.002)	**0.828** (-0.007)	0.817	0.817 (+0.000)	**0.808** (-0.008)
Economy	12/24/36	MSE	0.536	**0.530** (-0.006)	0.551 (+0.014)	0.303	**0.296** (-0.007)	0.328 (+0.025)
Economy	12/24/36	MAE	0.604	**0.601** (-0.003)	0.613 (+0.009)	0.435	**0.430** (-0.005)	0.456 (+0.021)
Energy	12/24/36/48	MSE	0.318	0.317 (-0.001)	**0.316** (-0.002)	0.254	0.253 (-0.001)	**0.248** (-0.006)
Energy	12/24/36/48	MAE	0.418	**0.416** (-0.001)	0.417 (-0.001)	0.366	0.362 (-0.005)	**0.357** (-0.009)
Public_Health	12/24/36/48	MSE	1.934	1.904 (-0.030)	**1.872** (-0.062)	1.589	**1.532** (-0.057)	1.536 (-0.053)
Public_Health	12/24/36/48	MAE	0.961	0.953 (-0.008)	**0.930** (-0.031)	0.836	0.822 (-0.013)	**0.809** (-0.027)
Security	12/24	MSE	73.195	**73.176** (-0.019)	73.205 (+0.010)	81.804	**80.005** (-1.799)	81.240 (-0.564)
Security	12/24	MAE	4.137	4.135 (-0.002)	**4.134** (-0.003)	4.996	**4.845** (-0.152)	4.945 (-0.051)
SocialGood	12/24/36/48	MSE	1.528	1.515 (-0.013)	**1.402** (-0.126)	1.515	1.505 (-0.010)	**1.341** (-0.173)
SocialGood	12/24/36/48	MAE	0.931	0.925 (-0.006)	**0.877** (-0.054)	0.791	0.786 (-0.005)	**0.732** (-0.058)
Traffic	12/24/36/48	MSE	0.294	**0.292** (-0.002)	0.295 (+0.002)	0.283	0.282 (-0.001)	**0.235** (-0.049)
Traffic	12/24/36/48	MAE	0.429	**0.422** (-0.007)	0.431 (+0.002)	0.348	0.346 (-0.002)	**0.312** (-0.036)

参考文献

[1] A. Zeng, M. Chen, L. Zhang, and Q. Xu. Are Transformers Effective for Time Series Forecasting? AAAI, 2023.

[2] Y. Nie, N. H. Nguyen, P. Sinthong, and J. Kalagnanam. A Time Series Is Worth 64 Words: Long-term Forecasting with Transformers. ICLR, 2023.

[3] H. Liu et al. Time-MMD: Multi-Domain Multimodal Dataset for Time Series Analysis. NeurIPS Datasets and Benchmarks, 2024.

[4] S. Wang, P. Chen, Y. Wang, W. Qiu, C. Guo, B. Yang, and Y. Shu. Unlocking the Value of Text: Event-Driven Reasoning and Multi-Level Alignment for Time Series Forecasting. ICLR, 2026.

[5] X. Wu, J. Jin, W. Qiu, P. Chen, Y. Shu, B. Yang, and C. Guo. Aurora: Towards Universal Generative Multimodal Time Series Forecasting. ICLR, 2026.

感谢老师们和助教们这学期的辛苦付出！

敬请老师们批评指正！

汇报人：10242150443 吴晨曦

时间：2026.6.26